

1. MAC Addresses: Standardization, Procedural Rules, and Historical Development

Standardisation Organisation and Family of Standards: The standardisation organisation responsible for defining and managing MAC addresses is the **Institute of Electrical and Electronics Engineers (IEEE)**. When describing MAC addresses in the context of wireless networks and mobile phones, the relevant family of standards is the **WiFi family of standards**, which is formally documented in the **IEEE 802.11** standard.

The Development of MAC Addresses Through the Years: To fully understand how MAC addresses come about, we must look at their extensively detailed development through the years and how they conform strictly to the procedural rules of the IEEE:

- **The Early Years (Standardizing the 48-bit structure):** The concept of the MAC address originated in the 1970s at Xerox PARC for early Ethernet networks. The IEEE formally standardized this by creating the EUI-48 (Extended Unique Identifier) format, consisting of 48 bits. According to the strict procedural rules of the IEEE Registration Authority, a MAC address comes about through a two-part collaborative process. First, a manufacturer must purchase a 24-bit Organizationally Unique Identifier (OUI) from the IEEE. Second, the manufacturer is procedurally responsible for assigning the remaining 24 bits to uniquely identify the exact piece of hardware they produce.
- **The Mobile & WiFi Era (The IEEE 802.11 integration):** During the late 1990s and 2000s, wireless technology boomed. The **IEEE 802.11** WiFi family of standards adopted these exact same 48-bit MAC addresses to identify wireless access points and mobile phones. However, this development inadvertently created a massive privacy vulnerability. As mobile phones became ubiquitous, they continuously broadcasted their hardcoded, permanent MAC addresses in "probe requests" to discover known WiFi networks, allowing service providers to track the physical movements of individuals over time.
- **Recent Years (The Shift to Privacy and Randomization):** In response to these severe privacy concerns, the development of MAC addresses underwent a revolutionary change starting around 2014. Modern mobile operating systems introduced **MAC Randomization**. Instead of continuously broadcasting the permanent, manufacturer-assigned MAC address, devices now dynamically generate random, temporary MAC addresses when scanning for WiFi networks. This conforms to the procedural rules of the IEEE standard by changing the "locally administered" bit to '1', signaling to the network that the address is a randomized identifier, thereby successfully balancing IEEE network standards with modern privacy protections.

2. Essay: Weighing Permanence vs. Privacy: A Legal Analysis of the Digital Object Architecture

Definition and Privacy Implications of DOA The Digital Object Architecture (DOA) is a framework that shifts the internet's routing mechanism from identifying *where* data is located (using dynamic IP addresses or URLs) to identifying *what* the data is. It achieves this by assigning a globally unique, persistent identifier to every "digital object" (e.g., a document or

dataset). The severe privacy implication of DOA is the creation of immutable digital footprints. Because the identifier never changes, any digital object containing personal data becomes permanently trackable across the internet, enabling effortless behavioral profiling by third parties.

Opinion: Advantages and Disadvantages In my opinion, DOA is a brilliant engineering concept for public information, but it is highly dangerous for personal data.

- **Advantages:** DOA effectively solves "link rot" (broken URLs) and enables seamless machine-to-machine interoperability. It ensures that valuable research or public records remain permanently accessible regardless of server hardware changes.
- **Disadvantages:** The primary disadvantage is that this permanence inherently centralizes tracking. Whoever manages the DOA resolution registry gains the ultimate surveillance capability to monitor exactly who is requesting which digital object globally.

Logical Reasoning and Legal Alignment My conclusion—that DOA is fundamentally incompatible with modern European privacy standards when applied to personal data—is reached through two logical steps grounded in the GDPR:

- **Step 1: The Conflict with Purpose Limitation.** The architectural core of DOA is to maximize data discoverability and reuse. However, **GDPR Article 5.1(b)** explicitly mandates the principle of *purpose limitation*, requiring that data not be further processed in a manner incompatible with its original intent. An architecture designed to make data universally available removes the technical friction needed to enforce these boundaries, directly facilitating unauthorized secondary processing.
- **Step 2: The Technical Impossibility of Erasure.** DOA is explicitly engineered for data permanence. This structurally contradicts **GDPR Article 17 (Right to Erasure / 'Right to be Forgotten')**. If a data subject requests the deletion of their personal data, an architecture that relies on immutable, persistent identifiers acts as a direct technical barrier to true erasure. Furthermore, **GDPR Recital (30)** warns that persistent online identifiers can be combined to create profiles of natural persons; DOA elevates this risk by making the data object itself the persistent identifier.

Conclusion Retracing these logical steps, we reach a clear conclusion: The very advantages of DOA (data permanence and interoperability) are the exact characteristics that violate the fundamental privacy rights (minimisation and erasure) guaranteed by the GDPR. Without strict *Privacy by Design* measures to anonymize these identifiers, DOA should not be utilized for personal data.